

```
import warnings
warnings.filterwarnings('ignore')
import pandas as pd
import re
import seaborn as sns
from sklearn.feature_extraction.text import TfidfVectorizer
import matplotlib.pyplot as plt
from wordcloud import WordCloud
```

```
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
```

 [nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.

```
data = pd.read_csv('flipkart_data.csv')
data.head()
```



	review	rating
0	It was nice produt. I like it's design a lot. ...	5
1	awesome sound....very pretty to see this nd th...	5
2	awesome sound quality. pros 7-8 hrs of battery...	4
3	I think it is such a good product not only as ...	5
4	awesome bass sound quality very good bettary l...	5

Next steps:

[Generate code with data](#)

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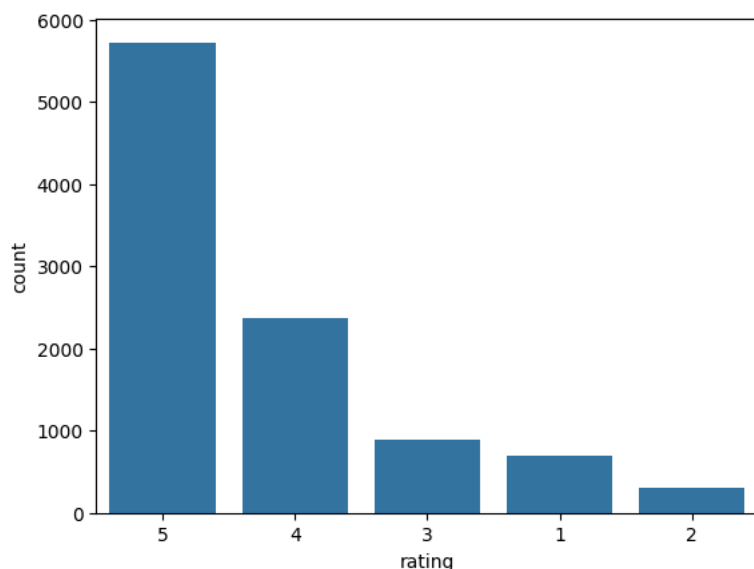
[New interactive sheet](#)

```
# unique ratings
pd.unique(data['rating'])
```

 array([5, 4, 1, 3, 2])

```
sns.countplot(data=data,
               x='rating',
               order=data.rating.value_counts().index)
```

 <Axes: xlabel='rating', ylabel='count'>



```
# rating label(final)
pos_neg = []
for i in range(len(data['rating'])):
    if data['rating'][i] >= 5:
        pos_neg.append(1)
    else:
        pos_neg.append(0)
```

```
data['label'] = pos_neg
```

```
nlk.download('punkt')
from tqdm import tqdm
```

```
def preprocess_text(text_data):
    preprocessed_text = []

    for sentence in tqdm(text_data):
        # Removing punctuations
        sentence = re.sub(r'^\w\s', '', sentence)

        # Converting lowercase and removing stopwords
        preprocessed_text.append(' '.join(token.lower()
                                           for token in nltk.word_tokenize(sentence)
                                           if token.lower() not in stopwords.words('english'))))

    return preprocessed_text
```

```
# This code is modified by Susobhan Akhuli
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
```

```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab') # Download the punkt_tab resource
nltk.data.find('tokenizers/punkt') # Ensure it is installed
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
FileSystemPathPrinter('/root/nltk_data/tokenizers/punkt')
```

```
data.head()
```

```

      review  rating  label
0  It was nice produt. I like it's design a lot. ...      5      1
1  awesome sound....very pretty to see this nd th...      5      1
2  awesome sound quality. pros 7-8 hrs of battery...      4      0
3  I think it is such a good product not only as ...      5      1
4  awesome bass sound quality very good bettary l...      5      1
```

Next steps: [Generate code with data](#) [View recommended plots](#) [New interactive sheet](#)

```
data["label"].value_counts()
```

```

      count
label
1      5726
0      4250
```

```
consolidated = ' '.join(
    word for word in data['review'][data['label'] == 1].astype(str))
wordCloud = WordCloud(width=1600, height=800,
                      random_state=21, max_font_size=110)
plt.figure(figsize=(15, 10))
plt.imshow(wordCloud.generate(consolidated), interpolation='bilinear')
plt.axis('off')
plt.show()
```



```
cv = TfidfVectorizer(max_features=2500)
X = cv.fit_transform(data['review']).toarray()
```

X

```
array([[0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       ...,
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.]])
```

[illegible]

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

```
model = DecisionTreeClassifier(random_state=0)
model.fit(X_train,y_train)
```

```
#testing the model
pred = model.predict(X_train)
print(accuracy_score(y_train,pred))
```

```
# This code is modified by Susobhan Akhuli
```

→ 0.9367050725721981

```
from sklearn import metrics
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_train,pred)
```

[illegible]

```
cm_display.plot()  
plt.show()
```

```
# This code is modified by Susobhan Akhuli
```

